SOFTWARE ENGINEERING

LAB-07

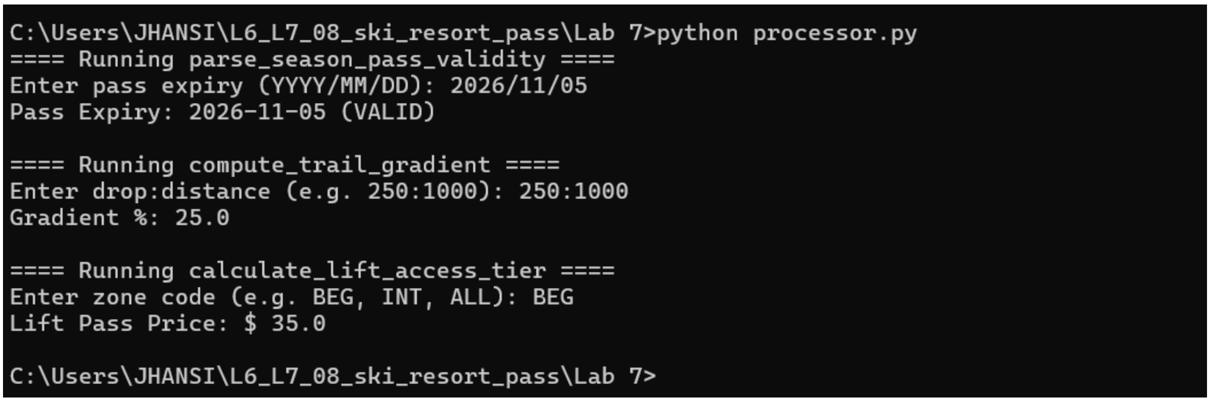
Fuzzing for Input Validation Bugs

NAME:MANDAGANDLA JHANSI

SRN:PES1UG24CS672

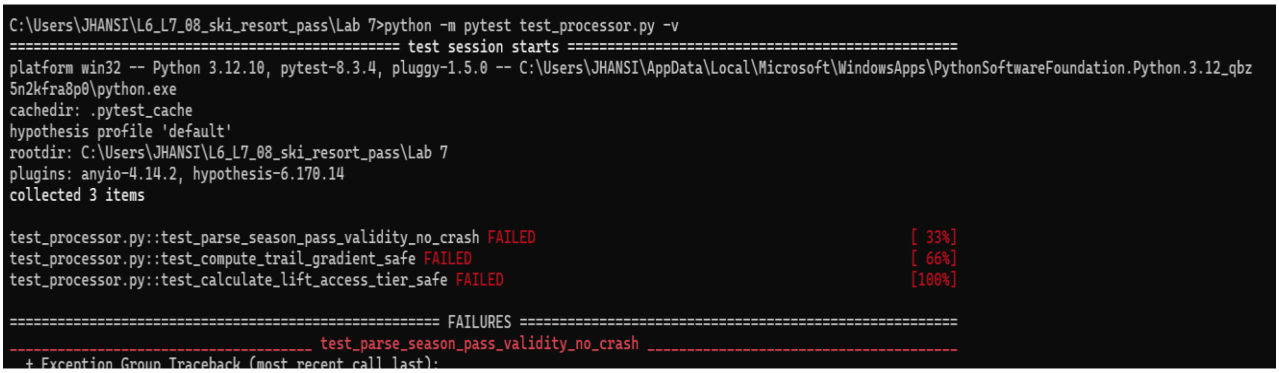
SECTION:L

Execution of the Original processor.py

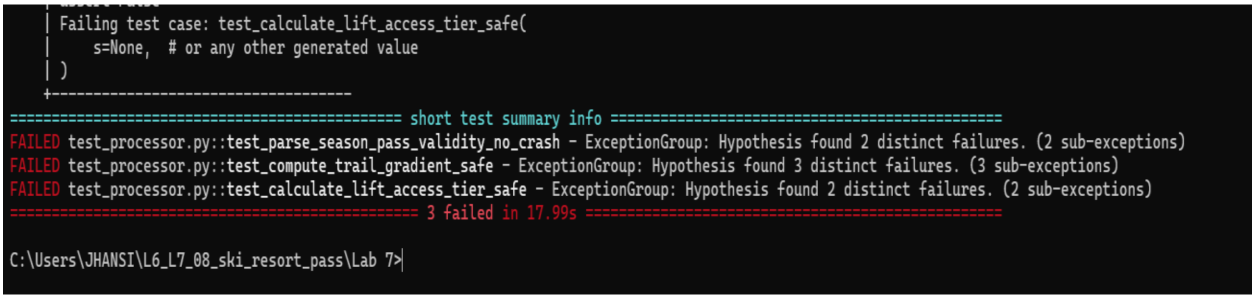


```
C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>python processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2026/11/05
Pass Expiry: 2026-11-05 (VALID)

==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 250:1000
Gradient %: 25.0

==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): BEG
Lift Pass Price: $ 35.0

C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>
```

Hypothesis-Based Testing – Initial Test Failures

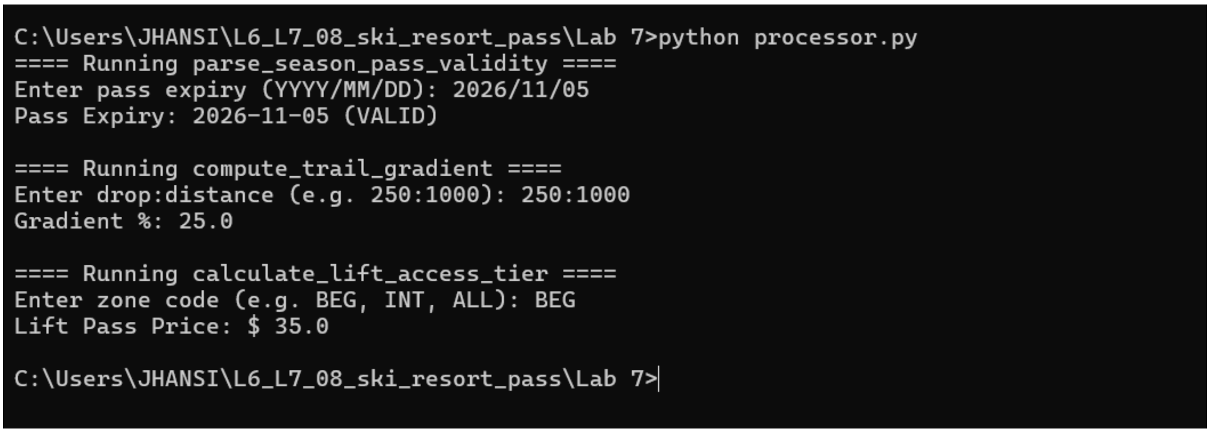


```
C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>python -m pytest test_processor.py -v
=============================================== test session starts ===============================================
platform win32 -- Python 3.12.10, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\JHANSI\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz
5n2kfra8p0\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7
plugins: anyio-4.14.2, hypothesis-6.170.14
collected 3 items

test_processor.py::test_parse_season_pass_validity_no_crash FAILED                                            [ 33%]
test_processor.py::test_compute_trail_gradient_safe FAILED                                                    [ 66%]
test_processor.py::test_calculate_lift_access_tier_safe FAILED                                                [100%]

==================================================== FAILURES =====================================================
____________________________________ test_parse_season_pass_validity_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
```



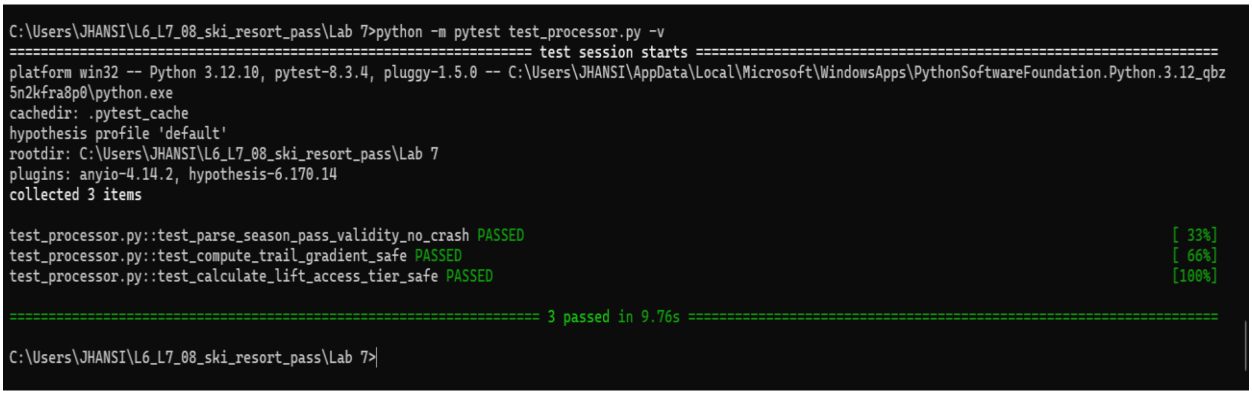
```
    | Failing test case: test_calculate_lift_access_tier_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
============================================= short test summary info =============================================
FAILED test_processor.py::test_parse_season_pass_validity_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_compute_trail_gradient_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_lift_access_tier_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
================================================ 3 failed in 17.99s ================================================

C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>
```

Execution of the Corrected processor.py



```
C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>python processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2026/11/05
Pass Expiry: 2026-11-05 (VALID)

==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 250:1000
Gradient %: 25.0

==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): BEG
Lift Pass Price: $ 35.0

C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>
```

Hypothesis-Based Testing – Successful Execution (All Three Tests Passed)



```
C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>python -m pytest test_processor.py -v
=================================================================== test session starts ===================================================================
platform win32 -- Python 3.12.10, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\JHANSI\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz
5n2kfra8p0\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7
plugins: anyio-4.14.2, hypothesis-6.170.14
collected 3 items

test_processor.py::test_parse_season_pass_validity_no_crash PASSED                                                                                  [ 33%]
test_processor.py::test_compute_trail_gradient_safe PASSED                                                                                          [ 66%]
test_processor.py::test_calculate_lift_access_tier_safe PASSED                                                                                      [100%]

==================================================================== 3 passed in 9.76s ====================================================================

C:\Users\JHANSI\L6_L7_08_ski_resort_pass\Lab 7>
```